$(x = \pm 2)$ $\left(\begin{array}{l} x^2 = \pm 54\,i \\ x^2 = \pm\sqrt{54}\,i \\ x = \pm 2i \end{array} \right)$

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chapter-1

Number System. $[i = \sqrt{-1}]$

→ Complex number (z)

general form $= \underbrace{a}_{\text{Real part}} + ib]$ → imaginary part

↳ iota

* imaginary no.

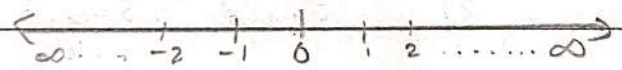
Eg - $2+i3$, 5, $2i]$ → Purely imaginary no.

$5 \to 5+0i$, $2i \to 2i+0$

Purely Real no.

Real numbers - (R)

• Numbers which can be represented on number line.



• Real no. are the combination of rational and irrational number.

→ Real numbers-

(a)

• Rational - The no's that can be written in the form of p/q, $q \neq 0$ and p, q are integers.

Eg- $\frac{1}{3}, \frac{2}{7}, \frac{5}{4}, \frac{-3}{2}$ etc.

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⇒ Eg → $\frac{5}{0} = x$

$5 = 0 \times x$

$5 = 0$ [Not true]

$x = 2 = 0 \times 2$

$5 = 0$ [Not true]

∴ Not defined (infinite)

Eg ⇒ $\frac{0}{0} = y$

$0 = 0 \times y$ (always true for any value of y)

⇒ Indeterminate form

↳ 7 form

4 algebraic

3 Exponential

Rational no. - (Ratio)

2.) the no. which has terminating Decimal expansion.

Eg → 1.34, 1.25, 2.345 etc.

3) the no. which has non-terminating repeating and decimal expansion

Eg → 3.41414141.... ($3.\overline{41}$)

$(a, b \in N$ Epsilon$)$

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(Q.T)

Irrational - the no's which can't be written in $\frac{p}{q}$ form.

- The numbers which have non terminating non repeating decimals expansion.

Eg → 1.0100100d1....

- square root of non-~~zero~~ perfect squares are irrational.

Eg → $\sqrt{2}$, $\sqrt{3}$ etc...

- Square root of prime numbers

Eg. $\sqrt{5}$, $\sqrt{7}$ ... etc.

Eg → π is irrational

$\frac{22}{7}$ is rational

- We use approx value of π as $\frac{22}{7}$

e → exponential → irrational

Rational no

↓

Fractions

- Part of something
- Fractions can't be negative
- $\frac{a}{b}$, a and b are natural no.

Types -

1. Proper = $\frac{a}{b}$, $b > a$

Eg → $\frac{3}{4}$, $\frac{1}{2}$ etc.

Value lies between 0 and 1.

2. Improper - $a > b$ $\left(\frac{a}{b}\right)$

Eg - $\frac{3}{2}$, $\frac{4}{3}$ etc

- Value is greater than 1.

Spiral

$\frac{7}{2}$

3. Mixed fraction-

- whole + fractional part
- $\underbrace{c}_{W} \frac{a}{b}$] - FP

Eg - $-1\frac{1}{2}$

4. Compound fraction

- $\dfrac{\frac{a}{b}}{\frac{c}{d}}$ → Normalize → upon symbol

$\Rightarrow \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$

Eg → $\dfrac{1/5}{3} = \frac{1}{5} \times \frac{1}{3} = \frac{1}{15}$

Eg → $2/3/7 = \frac{14}{3} \left(2 \times \frac{7}{3}\right)$

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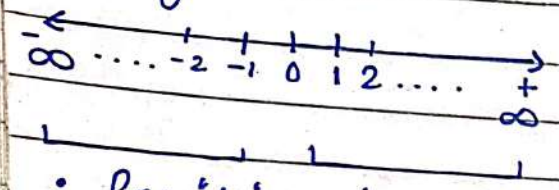
- Integers (z)



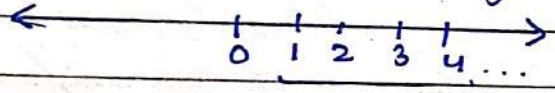
- Positive integer - 1, 2, 3, ...
- Negative integer - -1, -2, -3...
- Non-negative integer - 0, 1, 2...
- Non-~~neg~~ positive integer = 0, -1, -2...

* 0 is neigther negative not positive

- Whole no. (w)

0, 1, 2, 3...

"Alias → non-negative integer"



natural numbers (N)

Natural numbers

↓

- Even number
- Multiple of 2
- Eg → 2, 4, 6, 8, 10... 2n

↓

general form of even no.
- n is natural no.

- Odd number
- not multiple of 2
- Eg → 1, 3, 5, 7... (2n-1) → general form
n is a natural no.

Spiral

Smallest prime no. (3 digit) 101.

111 → 121

2, 3, 5, 7

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Prime no. –

1-100 = 25, 1-75 = 21, 1-50 = 15, 1-25 = 8

• Prime no. –

– no.'s which has only two factors

two factors
↓ ↓
1 itself

Eg → 5
1 5

5 - Prime

• Composite no.

– no.'s which has more than two factors

Eg. → 4
1 2 4

∴ 3 factor.
• 4 is a composite no.

• Note – • 2 is the smallest prime no.
• 1 is neither prime nor composite.

Co-prime number – The pair of numbers which has highest common factor equals to 1 (HCF = 1)

Eg → (3, 4), (2, 3), (4, 15)

• Twin Primes – The pair ~~in~~ of Prime numbers which has difference of (2)

Eg → (3, 5), (5, 7)

Spiral